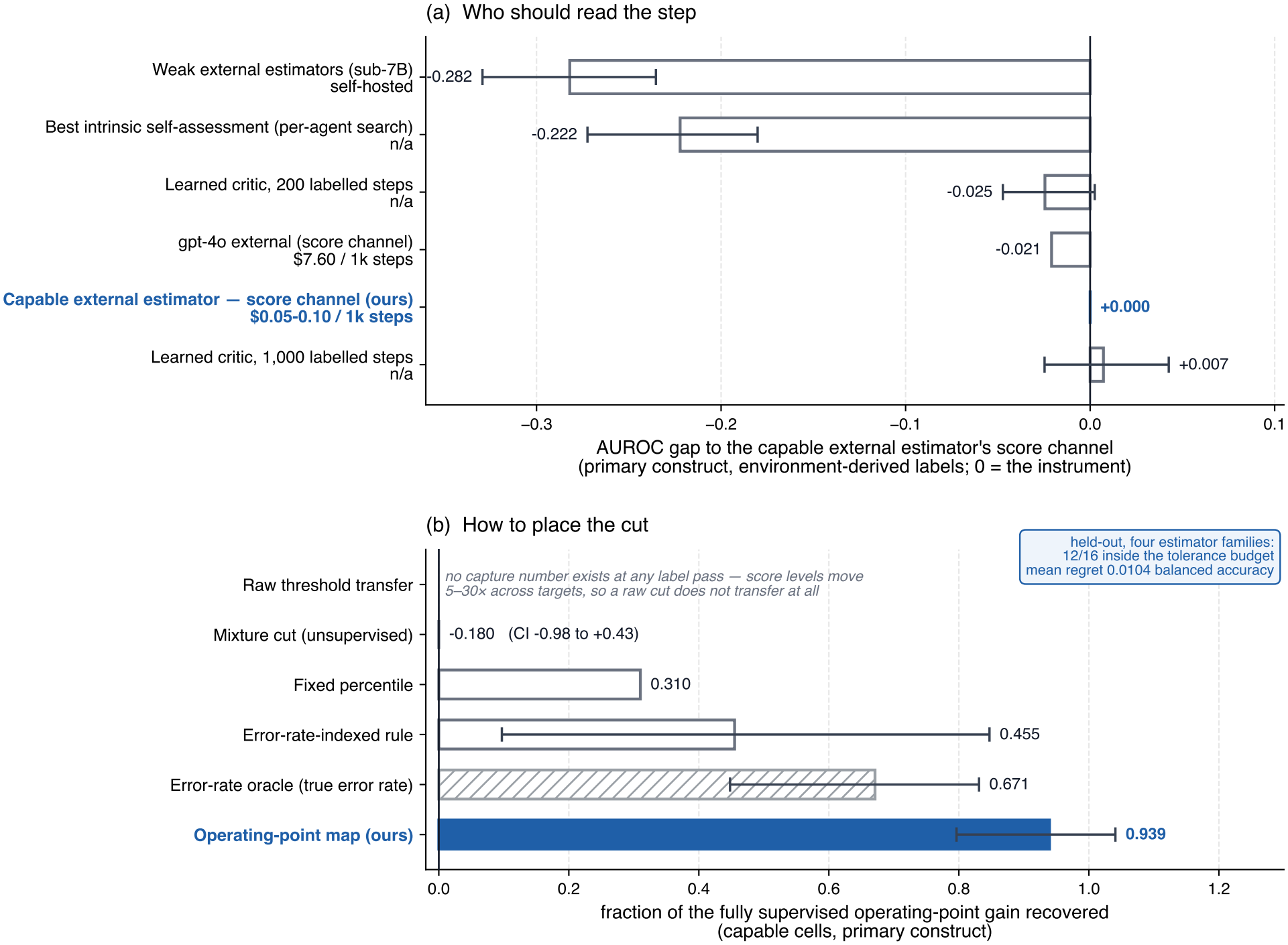
(a) Who should read the step
Weak external estimators (sub-7B)
self-hosted
-0.282
Best intrinsic self-assessment (per-agent search)
n/a
-0.222
Learned critic, 200 labelled steps
n/a
-0.025
gpt-4o external (score channel)
$7.60 / 1k steps
-0.021
Capable external estimator — score channel (ours)
$0.05-0.10 / 1k steps
+0.000
Learned critic, 1,000 labelled steps
n/a
+0.007
−0.3
−0.2
−0.1
0.0
0.1
AUROC gap to the capable external estimator's score channel
(primary construct, environment-derived labels; 0 = the instrument)
(b) How to place the cut
held-out, four estimator families:
12/16 inside the tolerance budget
mean regret 0.0104 balanced accuracy
Raw threshold transfer
no capture number exists at any label pass — score levels move
5–30× across targets, so a raw cut does not transfer at all
Mixture cut (unsupervised)
-0.180 (CI -0.98 to +0.43)
Fixed percentile
0.310
Error-rate-indexed rule
0.455
Error-rate oracle (true error rate)
0.671
Operating-point map (ours)
0.939
0.0
0.2
0.4
0.6
0.8
1.0
1.2
fraction of the fully supervised operating-point gain recovered
(capable cells, primary construct)